

Question	Answer	Marks
5(a)(i)	$P = 40$ and $R = 0$	A1
	$Q = 20$ and $S = -1$	A1
5(a)(ii)	(electron) antineutrino	B1
5(a)(iii)	leptons	B1
5(b)	(alpha particle consists of) 2 protons and 2 neutrons	C1
	quark composition of proton = up up down or quark composition of neutron = up down down	C1
	quark composition (of alpha particle) = 6 up, 6 down	A1

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